Homework 1

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1. [50%] A manufacturer determines that the profit P (in dollars) obtained by producing and selling x units of Product 1 and y units of Product 2 is approximated by the model

$$P(x, y) = 8x + 10y - (0.001)(x^2 + xy + y^2) - 10000.$$

Find the production level that produces a maximum profit. What is the maximum profit?

① Find partial derivative :

$P_x \triangleq \frac{\partial P}{\partial x} = 8 - 0.002X - 0.001y$

$P_y \triangleq \frac{\partial P}{\partial y} = 10 - 0.001X - 0.002y$

$\times 2$ subtract

$12 = 0.003y$

$y = 4000$

② Solve partial derivative $= 0$:

$8 - 0.002X - 4 = 0$

when $P_x = P_y = 0$, $\Rightarrow$ ($X = 2000$, $y = 4000$ (Critical point)

$4 = 0.002X$

$X = 2000$

③ Find second derivative of each terms in following:

$P_{xx} = -0.002$

$P_{xy} = -0.001$

$P_{yy} = -0.002$

④ Test the relative extrema of P based on quantity.

$d = P_{xx}(2000, 4000) \times P_{yy}(2000, 4000) - (P_{xy}(2000, 4000))^2$

$= (-0.002) \times (-0.002) - ((-0.001))^2 = 5 \times 10^{-6} > 0$

& $P_{xx}(2000, 4000) < 0$ $\Rightarrow$ P has a relative maximum at production level $(2000, 4000)$ (local maximum)

⑤ Apply $P(2000, 4000)$ to obtain maximum profit :

$P(2000, 4000) = 8 \times 2000 + 4000 \times 10 - (0.001)((2000)^2 + 2000 \times 4000 + (4000)^2) - 10000$

$= 16000 + 40000 - (0.001)(4000000 + 8,000,000 + 16,000,000) - 10,000 = 56,000 - 28,000 - 10,000 = 18,000$ #

(In dollars)

2. [50%] Determine whether the set of points satisfying a quadratic inequality:

$$C = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_2 \geq x_1^2\}$$

is convex. Provide a formal proof or a counterexample.

① Convert the inequality into a function form:

Let $f(x_1) = x_1^2$

Rewrite the quadratic inequality to $C = \{(x_1, x_2) \in R^2 \mid x_2 \geq f(x_1)\}$

② Take two points in the set to prove convexity:

Let $(x_1, x_2), (y_1, y_2) \in C$ & $(x_1, x_2) \neq (y_1, y_2)$

$\lambda \in [0, 1]$ to form convex combination with Internal Division Formula

$\Rightarrow (\lambda x_1 + (1-\lambda) y_1, \lambda x_2 + (1-\lambda) y_2)$

③ Apply Jesen's Inequality by f and convex combination:

Since $f''(x) = 2 \geq 0$, so $f(x)$ is a convex function

Because $f(x)$ is convex, so we can apply Jesen's Inequality

: $f(\lambda x_1 + (1-\lambda) y_2) \leq \lambda f(x_1) + (1-\lambda) f(y_2)$.

And since $x_2 \geq f(x_1)$, $y_2 \geq f(y_1)$.

we obtain:

$$\lambda x_2 + (1-\lambda) y_2 \geq \lambda f(x_1) + (1-\lambda) f(y_1) \geq f(\lambda x_1 + (1-\lambda y_1))$$

$$\therefore \lambda x_2 + (1-\lambda) y_2 \geq f(\lambda x_1 + (1-\lambda) y_1)$$

$$\therefore (\lambda x_2 + (1-\lambda) y_2, f(\lambda x_1 + (1-\lambda) y_1) \in C$$

$\Rightarrow$ C is a convex set